Research Log Book

Bhagyesh Rathi

Project: Knowledge Tracing for Large Language Models

CMPE 295A, San José State University, Fall 2026 · Advisor: Dr. Mahima Suresh Agumbe

Team: Ronak Patel, Bhagyesh Rathi, Aniket Sarrin · Repository: github.com/bhagyeshrathi07/LLM-Knowledge-Tracing

Period covered: August 31 – October 29, 2026 · Prepared October 29, 2026

Role and track

Learner models (track owner since Sep 29): Bayesian Knowledge Tracing; design-decision write-ups; repository owner and advisor liaison.

Summary of contribution to date

- Prepared the scope questions for the Sep 1 advisor meeting; wrote the abstract's background and gap paragraphs and the formatted abstract submission; created the shared GitHub repository.
- Carried out an early literature survey (Sep 2–3) with a first project plan; read and summarised the classic and extended BKT literature; wrote up Design Decisions 1 and 2 and prepared Decisions 3 to 5 for adoption.
- Prepared the abstract presentation slides and script and delivered slides 1 to 6 of the Workbook 1 presentation; advisor liaison (recurring-slot request, Oct 19 meeting).
- Implemented Bayesian Knowledge Tracing for batched checkpoints: log-space forward-backward, MAP-EM with shrunk recency-weighted per-item guess and slip, two anchors, mastery curves; ran the 61-setting grid and the horizon-5 and -20 runs; wrote the design spec, plan, results tables and figures (14 commits in PR #5, merged Oct 27; the README learner-model section is on bhagyesh_branch pending its own pull request).
- Refreshed the team briefing and project notes (status tables, chain-of-thought future-work section, README layout).

Weekly log

Each entry lists the meetings attended, the work done that week by this member (as opposed to the team), and the artifacts that evidence it (repository paths, commits, documents).

Week 1: Aug 31 – Sep 6

Meetings	Aug 31 advisor meeting (Google Meet). Sep 1 impromptu advisor meeting (in person; note-taker).
Work done	• Took part in choosing the project from the advisor's proposed ideas and refining the framing. • Prepared the list of scope questions and the request for papers the advisor has reviewed, for the Sep 1 meeting (action item, due Sep 1). • Sep 2: drafted abstract paragraphs 1 and 2 (background and gap statement); built the formatted abstract document (title page and body per the course template). • Sep 2–3: early literature survey over arXiv, OpenReview and recent venue proceedings with each recent claim fact-checked against primary sources; produced the brief 'Tracing What LLMs Know', a 115-entry reference list and a first project plan (research questions, baseline tiers, forward-chained held-out next-checkpoint evaluation, item-parameter drift test, simulation study, two-semester timeline) reviewed against the NeurIPS checklist (as recorded in the submitted Workbook 1, §8.6 and §9.1). • Sep 3: created the shared GitHub repository (initial commit, licence, ignore rules) and the shared artifacts folder for course submissions.
Artifacts	meetings/2026-09-01-advisor.md; Project_Abstract_KT_for_LLMs.pdf; repository bhagyeshrathi07/LLM-Knowledge-Tracing; brief 'Tracing What LLMs Know' and the first project plan (shared folder).

Week 2: Sep 7 – Sep 13

Meetings	Sep 8 team meeting (Discord).
Work done	• Reviewed the advisor's abstract feedback with the team. • Took the classic and extended BKT reading: Corbett & Anderson 1995, Khajah, Lindsey & Mozer 2016, pyBKT (Badrinath et al. 2021); wrote one-paragraph summaries. • E-mailed the advisor to request a recurring bi-weekly meeting slot (action item, due Sep 12).
Artifacts	meetings/2026-09-08-team.md; paper summaries.

Week 3: Sep 14 – Sep 20

Meetings	Sep 15 team meeting (Discord).
Work done	• Presented the BKT summaries: a two-state HMM per skill with learn, guess, slip and optionally forget; fits with very few learners, which is why it suits six models. • Took the action item to write up Decision 1 (time step = checkpoint) and Decision 2 (model family given few learners) as a pros-and-cons document (due Sep 22).
Artifacts	meetings/2026-09-15-team.md.

Week 4: Sep 21 – Sep 27

Meetings	Sep 23 team meeting (Discord).
Work done	• Delivered the Decisions 1 and 2 write-up with pros and cons; reviewed the literature-search result with the team. • Took the action item to prepare Decisions 3 to 5 (Phase 1 data source, skill tags, pseudo-learners) for the Sep 29 meeting. • Sep 26: completed the abstract presentation slides and speaker script (as recorded in the submitted Workbook 1, §8.6).
Artifacts	meetings/2026-09-23-team.md; Decisions 1–2 write-up (folded into docs/project-notes.md §6); abstract presentation slides.

Week 5: Sep 28 – Oct 4

Meetings	Sep 29 team meeting (Discord; note-taker). Workbook 1 submitted Oct 1.
Work done	• Presented Decisions 3 to 5 for adoption; the team adopted Decisions 1 to 5 and the three-track split (Bhagyesh: learner models). • Sep 30: set up the development environment on a fresh clone (Python 3.12 virtual environment, 52 tests passing, ruff clean); Node.js (docx, pptxgenjs) used for the abstract document and presentation. • Sep 30: commit b2c5717: moved project notes and abstract v2 into docs/, made the scripts executable, added .DS_Store to .gitignore. • Wrote the individual-contribution paragraph and updated the Chapter 1 search description, the industry paragraph, and the Chapter 8 and 9 entries in the submitted Workbook 1. • Took the Sprint 3 action items: read pyBKT's internal data format and the emIRT::dynIRT documentation and plan the BKT and dynIRT fits (due Oct 7); implement T-SKIRT (due Oct 14).
Artifacts	meetings/2026-09-29-team.md; commit b2c5717; docs/workbook/Workbook1.pdf (submitted version).

Week 6: Oct 5 – Oct 11

Meetings	Oct 9: Workbook 1 presentation delivered in class (slides 1–6).
Work done	• Studied pyBKT's E-step (EM_fit.py) and the emIRT::dynIRT data model. Finding recorded in docs/project-notes.md: pyBKT multiplies raw emission probabilities before normalising and underflows for 100 to 1,500 items per checkpoint, so a log-space implementation is required; dynIRT assigns each item to one period, so tied item parameters need a custom model. • Prepared and delivered slides 1 to 6 of the Workbook 1 presentation on Oct 9 (title, one-score-hides-four-stories, three literatures, feasibility, the bar, learner models under study). • T-SKIRT (assigned Oct 14) was implemented on the infrastructure track in the same sprint; the BKT plan from this study became the Oct 27 implementation.
Artifacts	docs/project-notes.md §4 item 5 (pyBKT and dynIRT findings); Workbook1_Presentation.pptx.

Week 7: Oct 12 – Oct 18

Meetings	None recorded. Advisor meeting booked for Oct 19.
Work done	• As advisor liaison, arranged the Oct 19 advisor meeting (first since the Sep 3 e-mail feedback).
Artifacts	Workbook1_Presentation.pptx.

Week 8: Oct 19 – Oct 25

Meetings	Oct 19 advisor meeting (11:20–11:40 PST).
Work done	• Attended the advisor meeting where the results to date were presented; took the action items: log-space BKT with forgetting and per-item emissions scored against the item-mean bar (due Nov 2), and the chain-of-thought / hallucination direction written into the project notes as future work (due Oct 26). • Designed the BKT implementation: per-skill two-state HMM with one transition per checkpoint, batched log-space emissions, per-item guess and slip shrunk toward an anchor (skill-shared or Fluid 2PL) with recency weighting.
Artifacts	docs/superpowers/specs/2026-10-27-bkt-design.md; docs/superpowers/plans/2026-10-27-bkt.md.

Week 9: Oct 26 – Oct 29

Meetings	No meeting. PR #5 merged Oct 27.
Work done	• Oct 27: implemented src/llm_kt/bkt.py: batch_log_emissions and log_forward_backward (tested against brute-force enumeration and a 2,000-item checkpoint for underflow), MAP-EM per skill with shrunk recency-weighted per-item guess/slip, BKT predictor with per-skill cache, mastery curves, skill and Fluid anchors; 26 tests. • Added ItemMeanEWMA (recency-weighted item mean) and SkillMean to src/llm_kt/baselines.py with tests; extended scripts/run_models.py with the bkt, item_mean_ewma and skill_mean predictors and the kappa / rho / anchor / theta flags. • Ran the 61-setting BKT grid at h = 1 (about 10 min per setting, eight parallel chunks) and the chosen settings at h = 5 and 20 (about 70 min per setting) on OLMo-2 MMLU; wrote scripts/analyze_bkt.py (selection on validation, comparison with the item mean and the control, four figures, tables under docs/results/bkt/). Result: BKT beats the bar at every horizon (test log-loss 0.310 vs 0.375 at h = 1) and ties the recency-weighted item mean. • Wrote docs/project-notes.md §9 (BKT design, why, results), §8 (chain-of-thought future work), the status table and §5.8 in docs/team-briefing.md, the BKT design spec and implementation plan (docs/superpowers/); made the five newer scripts executable. 14 commits in PR #5, merged Oct 27. The README learner-model section and results table (7f6c496) was committed after the merge and sits on bhagyesh_branch pending its own pull request.
Artifacts	Commits 00e9a6f … b763c72 (PR #5); 7f6c496 (bhagyesh_branch, unmerged); src/llm_kt/bkt.py; tests/test_bkt.py, test_baselines.py, test_analyze_bkt.py; scripts/analyze_bkt.py; docs/results/bkt/*; docs/figures/bkt_*.png.

Signed: _____ Date: _____